



Research Paper

A solvent-targeted recovery and precipitation scheme for the recycling of up to ten polymers from post-industrial mixed plastic waste

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ABSTRACT

Solvent-Targeted Recovery and Precipitation (STRAPTM) separates polymers within a plastic waste stream by selective dissolution. In this work, the STRAP framework, which combines computational modeling and experiments, was applied to develop a series of steps to separate up to 10 polymers from post-industrial mixed plastic waste (MPW) and the main components recovered were LDPE, HDPE, and PET. The STRAP steps were initially demonstrated with a physical polymer mixture containing LDPE, HDPE, PS, PVC, EVOH, PET, PP, PA6, PA66, and PA66/6, in which recoveries of 89% or higher were achieved for each polymer. This paper demonstrates a solvent selection approach that can be applied to separate unknown plastics materials into purer components.

1. Introduction

One of the challenges with plastic recycling is the processing of mixed plastic waste (MPW) and proper sorting of its components can dictate the efficiency of the overall recycling process (Ragaert et al., 2020). Plastics that are present in an MPW feedstock can have distinct properties and require different recycling approaches (Thiounn and Smith, 2020). The presence of multiple plastics, additives, and contaminants is a significant concern with mechanical recycling, for which waste stream purity is an important factor (Billiet and Trenor, 2020). In mechanical recycling, there are separation and sorting steps, bailing, washing, grinding, compounding, and pelletizing. The plastics can be separated in various ways by density, size, color, shape, or chemical composition (Ragaert et al., 2017). One disadvantage with mechanical recycling is the degradation of polymer properties, which means that the MPW is converted to less valuable products (Jeswani et al., 2021). Another disadvantage of mechanical recycling is that additives or contaminants are not removed and stay inside the recycled plastic. Compatibilization offers a solution to mechanically recycle multiple polymers, however, this becomes more challenging with complex polymer mixtures (Fortelný et al., 2004).

Other technologies do not require extensive sorting of plastic waste.

For example, in a pyrolysis approach where plastics are processed at elevated temperatures in the absence of oxygen, different plastics can be included in the feed to produce valuable oils, waxes, and gases that can be converted to chemicals and fuels (Jeswani et al., 2021; Kremer et al., 2021; Martínez-Narro et al., 2022). Different polymer mixtures containing high-density polyethylene (HDPE), polypropylene (PP), polyethylene terephthalate (PET), and polystyrene (PS) have been successfully pyrolyzed (Chattopadhyay et al., 2016). This approach offers advantages since it can process mixtures of plastics, including multilayer packaging, which cannot be processed with mechanical recycling (Dahlbo et al., 2018). A disadvantage of thermally converting plastics is that the process produces an oil that must be further converted (Billiet and Trenor, 2020; Schwarz et al., 2021). Feedstocks that contain polyvinyl chloride (PVC) do become a concern with pyrolysis due to the release of HCl which is toxic and can corrode the processing units (Brebú et al., 2004; Lopez-Uriónabarrenechea et al., 2012; Ragaert et al., 2020).

Other chemical conversion approaches for mixed plastics have been disclosed in the literature, including chemical oxidation and bioconversion (Sullivan et al., 2022), hydrogenolysis (Rorrer et al., 2021), gasification (Dogu et al., 2021; Wilk and Hofbauer, 2013), one-pot depolymerization (Yang et al., 2022), three-step depolymerization (Spicer et al., 2024). These alternative chemical approaches present

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viable pathways to process mixed waste; however, tuning conditions for each component and ensuring selectivity for the desired products can be a challenge.

Dissolution-based recycling methods offer the recovery of single polymer components from plastic waste (Miller-Chou and Koenig, 2003). In these methods, solvents are employed to selectively separate materials of interest from a single stream. Many solvents have been disclosed in the literature and in patents for most of the common polymers found in plastics produced today (Li et al., 2022). These dissolution-based processes do not break down the polymer chains and have shown advantages over chemical recycling in terms of energy requirements and potential environmental impacts (Billiet and Trenor, 2020; Schwarz et al., 2021). Solvent-Targeted Recovery and Precipitation (STRAPTM) is one dissolution-based approach that can process different types of plastic waste, including multilayer films and disposable facemasks, guided by thermodynamic calculations of polymer solubility (Sánchez-Rivera et al., 2023; Sánchez-Rivera et al., 2021; Walker et al., 2020; Yu et al., 2023; Zhou et al., 2021). STRAP has been demonstrated to recover up to 5 different polymeric components from a single, clean, post-industrial waste stream. The materials contained polymers like polyethylene (PE), ethylene vinyl alcohol (EVOH), polyethylene terephthalate (PET), ethylene vinyl acetate (EVA), and glycol-modified polyethylene terephthalate (PETG). Polymer separation by dissolution can be an effective way of processing MPW if selective solvents can be found for each polymer of interest, and moreover, can be applied to separate multilayer films that cannot be recycled mechanically.

In this work, STRAP was applied to recover different constituent polymeric materials from an unknown post-industrial MPW stream. Our computational and experimental tools allowed us to develop a series of steps to selectively separate 10 different polymers by selective dissolution, both with physical mixtures of pure polymer pellets and MPW. The polymers considered in this study were LDPE, HDPE, PS, PVC, EVOH, PET, PP, polyamide 6 (PA6), PA66, and PA66/6. The application of the selected solvents can recover valuable plastic materials from MPW, which can serve as a pretreatment for other recycling technologies or to determine which components are present in majority within the waste.

2. Materials and methods

2.1. Materials

The resins used in this study were: LDPE DOW 608A, EVOH EV3251, PET DAK 9921, PA6 B36, PA66 Zytel FG42A, and PA66/6 Ultramid C40L. The remaining polymers were received from Sigma-Aldrich: PP 427888, HDPE 427985, PS 182427, and PVC 389323. The solvents in this study were also received from Sigma-Aldrich: toluene (ACS reagent, $\geq 99.5\%$), DMSO (ACS reagent, $\geq 99.9\%$), gamma-valerolactone (ReagentPlus®, $\geq 99\%$), tetrahydrofuran (THF) (ACS reagent, $\geq 99.0\%$), 1,2-propanediol ($\geq 99\%$ for analysis from Aeros Organics), formic acid (ACS reagent reagent, Ph. Eur., $\geq 98\%$), and o-xylene (reagent grade, 98.0%). The post-industrial mixed plastic waste (MPW) feed was collected from different plastic packaging manufacturers by Convergen Energy, Wisconsin (CEW). The MPW was processed by CEW; first, it was sorted from metal, glass and stone; and second, it was shredded by a three-stage shredding process, to 10-in, then to 2–4-in, then < 0.5 -in. The same material was shipped to Michigan Technical University, where it was further shredded by a cross-cut shredder (make: Allegheny 16-75CX) until reaching a flake size of 2–3 mm with an aspect ratio of 1.

2.2. Experimental procedure

A 250 mL round bottom flask connected to a reflux condenser with a cold-water supply line was used for the STRAP experiments. The round bottom flask, which would contain the corresponding solvent and solids, was partially submerged in a 1500 mL dish containing silicone oil as a

heat transfer fluid. The system was heated to the desired dissolution temperature with an electric heat plate equipped with a magnetic stir drive and the stirring rate was adjusted to maintain constant mixing. We experimentally assessed the recovery of each polymer from the feed by performing the following processing steps: (1) dissolving the specific polymer in the selected solvent, then separating the solubilized polymer from the rest of the polymers via mechanical filtration, (2) reducing the solvent temperature and/or adding an antisolvent to precipitate the dissolved polymer, and (3) separating the precipitated polymer from the solvent via mechanical filtration. These steps were repeated for each of the targeted polymers in the mixed plastic waste. The recovered polymers were dried in a vacuum oven for 1 h at $100\text{ }^{\circ}\text{C}$ to remove residual solvents.

2.3. Computational methods

We evaluated temperature-dependent polymer solubilities computationally following the approach developed in our previous work. Specifically, we used the conductor-like screening model for realistic solvents (COSMO-RS), a combined quantum chemical and statistical mechanical approach, to calculate temperature-dependent polymer solubilities in various solvent systems (Klamt, 1995; Klamt et al., 1998; Zhou et al., 2021; Zhou et al., 2023). As shown in Fig. 1, COSMO-RS represents each molecule based on the screening charge density that arises at molecular surfaces due to polarization of the medium. Screening charge density profiles were obtained from density functional theory calculations in our previous work (Zhou et al., 2023). We approximated the profiles of polymer molecules by oligomer structures with deactivated terminal groups (Kahlen et al., 2010). COSMO-RS solubility calculations were performed by COSMOtherm 19 with the parameterization BP_TZVP_19 (COSMOlogic GmbH & Co KG, 2019; Klamt et al., 1998). The workflow established in our previous work can be adapted by using different reference inputs so that resin-specific polymer properties can be addressed with a minimum number of calculations. In this work, HDPE calculations use the PE molecular model with a reference input of 16.2 wt% in dodecane at $120\text{ }^{\circ}\text{C}$. PA66/6 calculations use the PA66 model with a reference input of 9.1 wt% in DMSO at $135\text{ }^{\circ}\text{C}$. Note that PA66/6 is a random copolymer with typical composition of 90 % PA 66 and 10 % PA 6, thus we employed the PA66 model as an approximation (Page, 2000). Calculations for all other polymers use the same parameters as our previous work (Zhou et al., 2023).

2.4. Characterization methods

Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) was used to characterize the MPW feedstock and the separated polymer fractions from the STRAP process and compare their spectra to virgin resins. The instrument was a Nicolet TM iS50 spectrometer with a liquid nitrogen cooled MCT detector. In a typical measurement, 128 scans were averaged with a 4 cm^{-1} resolution and range from 4000 to 400 cm^{-1} . The thermal properties of the recovered polymer fractions were analyzed with a TA instruments Q100 differential scanning calorimeter (DSC). Optical images of the MPW were taken with a Leica DM4 microscope.

3. Results

3.1. Characterization of the mixed plastic waste (MPW) feed

The MPW in this study was collected and shredded from different plastic packaging manufacturers. Plastic, metal, and cardboard components were in the feed as shown in Fig. 2(a). An initial analysis was done with infrared spectroscopy (FTIR) to understand which polymers were present in the post-industrial MPW. As seen in Fig. 2(b–d), the feedstock contained PE, PET, and polyamides (PA) from three separate

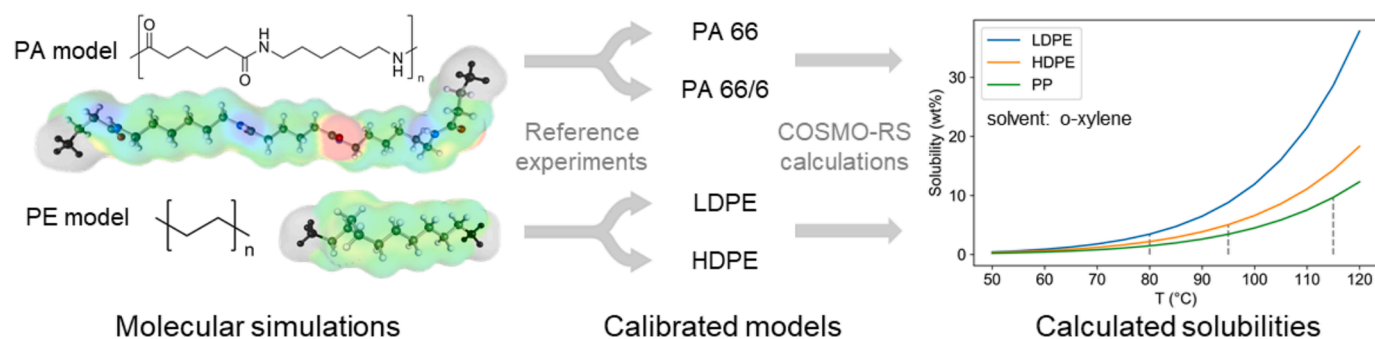


Fig. 1. Computational methods for polymer solubility prediction. Left: example COSMO-RS screening charge densities (colored surfaces) of PE and PA models. Oligomer molecules with end groups neglected (gray surface) are used to represent longer polymer chains. Middle: the same molecular model can be calibrated with different reference experimental data to represent different materials (e.g. PE model for LDPE and HDPE, PA model for PA 6 and PA 66). Right: example calculation results for temperature-dependent solubilities of LDPE, HDPE, and PP in o-xylene. Dashed lines refer to selected temperatures in experiments.

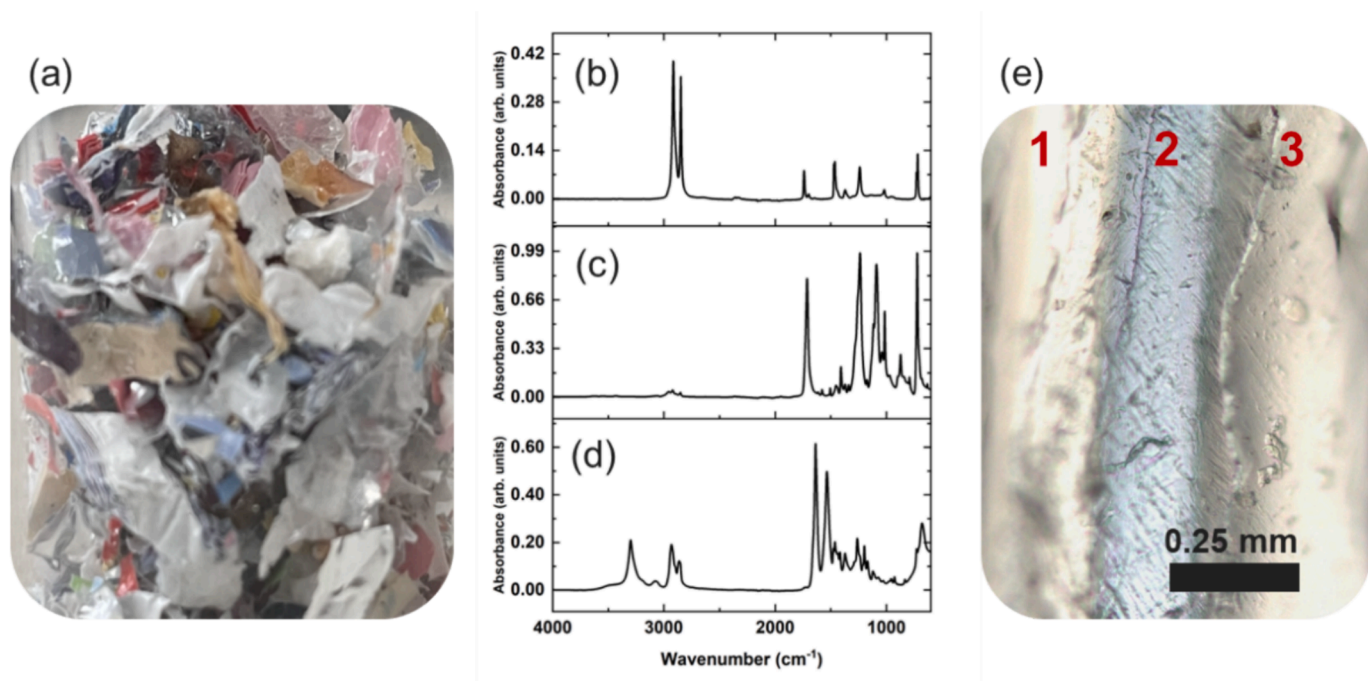


Fig. 2. Characterization of the post-industrial mixed plastic waste (MPW). (a) MPW feed, (b) FTIR of one sample indicating PE, (c) FTIR of one sample indicating PET, (d) FTIR of one sample indicating PA, and (e) optical microscopy image of one sample indicating the presence of multilayer plastic components. FTIR spectra of reference materials are shown in Fig. 3.

measurements (reference FTIR spectra for individual resins used to make this comparison are shown in Fig. 3). Since the MPW was collected from multiple manufacturers in the packaging industry, the presence of multilayer films was considered. Optical microscopy images were taken of the plastic components within the MPW and provided evidence of multilayer films (Fig. 2(e)). The presence of multilayer films gave indication that some of the components might not be separated by only using mechanical methods, and thus an alternative would be needed for their recovery and recycling. We hypothesized that the recovery of the plastic components within the MPW could be achieved by selective polymer dissolution. Since the actual composition of the MPW was unknown, a model physical mixture containing 10 common packaging polymers was used to develop each STRAP step. The virgin polymers in the physical mixture were LDPE, HDPE, PS, PVC, EVOH, PET, PP, PA6, PA66, PA66/6, PS, and PVC. Our computational tools allowed us to predict solubility values for each polymer in solvents of interest; solvents that were already disclosed in the literature for each polymer were also considered (Li et al., 2022).

3.2. Developing the STRAP steps with a polymer physical mixture

COSMO-RS calculation results and experimental demonstrations were used to identify potential solvent candidates for selective polymer dissolutions in each step and to determine the separation sequence of the ten polymers. The primary goal of solvent selection in each step was to obtain a higher solubility of the target polymer in the selected solvent than all other polymers, enabling selective dissolution. Table 1 summarizes the steps, selected solvents, and the COSMO-RS predicted solubilities of both the target polymer and other polymers for each step. As our previous study of the computational models indicated that predicted solubilities should be tested experimentally due to potential resin-specific differences (Zhou et al., 2023), a set of experiments (Tables S1 and S2) were conducted to validate the prediction data and finalize the STRAP process.

In general, polymers like PS and PVC can be dissolved at relatively low temperatures in different solvents (Achilias et al., 2009; Kampouris et al., 1987, 1988; Nauman and Lynch, 1993). For this reason, these two

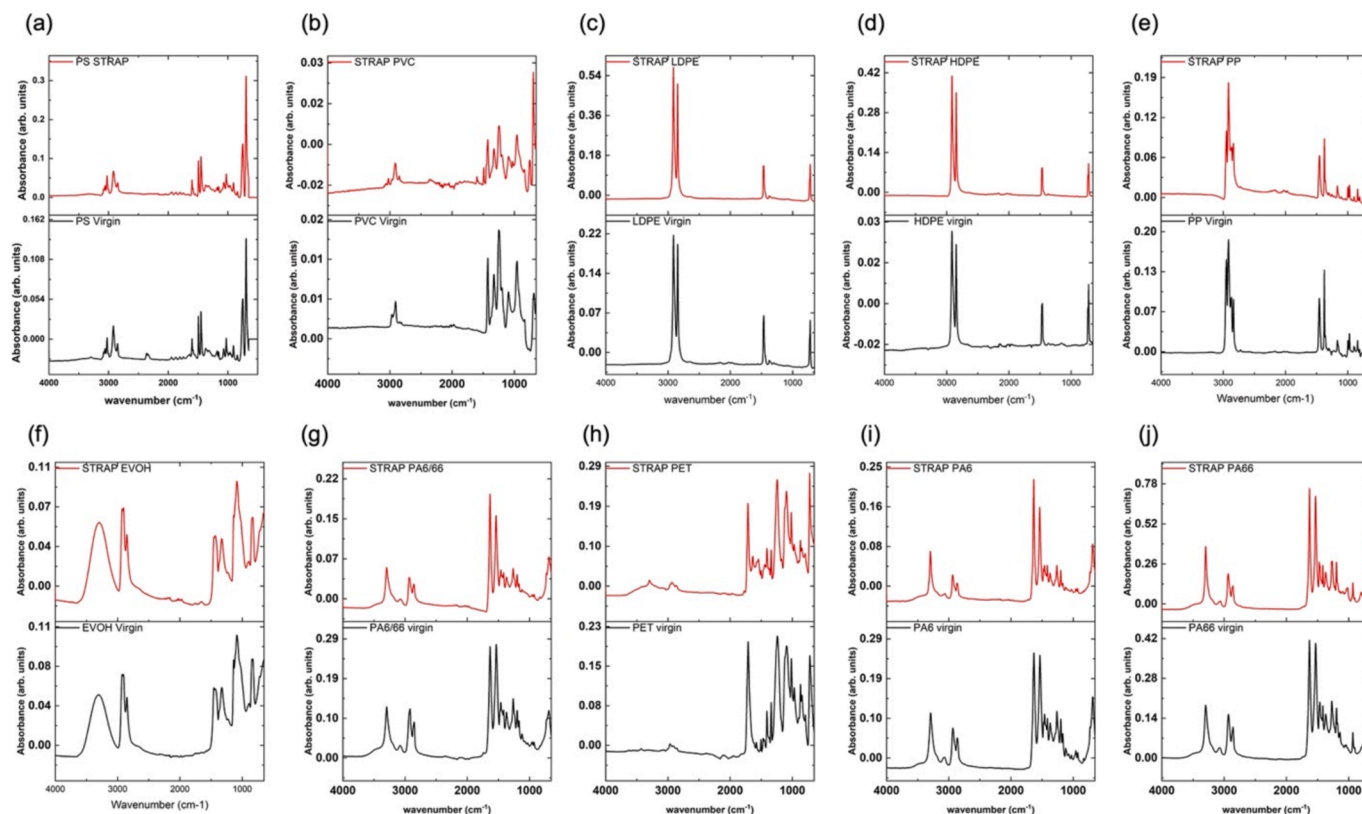


Fig. 3. IR spectra of the recovered STRAP polymers from the physical mixture (in red) and the virgin polymers (in black): (a) PS, (b) PVC, (c) LDPE, (d) HDPE, (e) PP, (f) EVOH, (g) PA66/6, (h) PET, (i) PA6, and (j) PA66. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

COSMO-RS predicted polymer solubilities for each STRAP step at the specified temperature.

Step	Solvent	T (°C)	Polymers and their solubilities (unit: wt%, bolded values are target polymers)									
			PS	PVC	LDPE	HDPE	PP	EVOH	PA66/6	PET	PA6	PA66
1	toluene	35	5.72	1.12	0.13	0.11	1.74	0.00	0.01	0.01	0.00	0.00
2	THF	67		19.10	2.09	1.45	2.88	1.12	1.33	0.99	0.52	0.26
3	o-xylene	80			3.43	2.17	1.46	0.01	0.14	0.18	0.12	0.03
4	o-xylene	95				5.04	3.42	0.04	0.39	0.53	0.35	0.09
5	o-xylene	115					9.65	0.33	1.29	1.88	1.23	0.33
6	DMSO/water	95						7.67	0.03	0.00	0.01	0.00
7	1,2-PDO	125							3.35	0.05	2.17	0.92
8	GVL	160								12.45	7.71	4.14
9	DMSO	145									8.41	3.56
10	formic acid	90										16.90

polymers were selected for the first two steps in STRAP with the 10-component physical mixture. For example, toluene, tetrahydrofuran (THF), and styrene have been previously identified as good solvents for PS (Zhou et al., 2023). Solubility calculations in these solvents show that toluene has excellent selectivity for PS at a low temperature. Therefore, toluene at 35 °C was selected as the solvent for PS, and this dissolution was placed as the first in the separation process. Since only PVC was predicted to have high solubility in THF at its boiling temperature, THF was selected as the solvent for the second step (Table 1). Following these first steps, one difficult separation to achieve with dissolution is between polyolefins like PE and PP due to their similar structures. In previous studies, PE and PP have been effectively separated by chromatographic methods and gravity separation and ozonation (Macko and Pasch, 2009; Reddy et al., 2011). Normally, when it comes to separation by dissolution, the same solvents can dissolve both polymers and different temperatures can be used to achieve the desired separation. For example, p-cymene can dissolve both PE and PP at 95 and 120 °C, respectively.

Other solvents like toluene, xylene, and THF have been disclosed to be selective for PP, LDPE, and HDPE at different temperatures, ranging from 50 to 160 °C (Nauman and Lynch, 1991). In our 10-step system, o-xylene was selected for the sequential dissolution of LDPE, HDPE, and PP at different temperatures. According to our solubility predictions, o-xylene can target LDPE, HDPE, and PP at 80, 95, and 115 °C, respectively, in steps 3, 4, and 5 (Table 1). After the dissolution steps for the polyolefins, in step 6, EVOH was dissolved in a 60 % DMSO-40 % water (v/v) mixture, which has been used effectively for STRAP with other materials like rigid and flexible multilayer packaging films in the recovery of EVOH (Sánchez-Rivera et al., 2023; Sánchez-Rivera et al., 2021).

Another difficult separation is for the three PAs in the physical mixture: PA6, PA66, and PA66/6. Common solvents that have been used for the dissolution of PAs are DMSO and formic acid, along with diols (Kartalis et al., 2002; Subramanian, 1995). To support our solubility predictions, these solvents were tested experimentally to observe

potential dissolutions and determine the proper steps to dissolve each PA separately. As seen in Table S1, PA6 was only soluble in DMSO and formic acid, at 145 °C and 60 °C, respectively. PA66 was only soluble in formic acid at 60 °C, and PA 66/6 was soluble in DMSO at 145 °C, 1,2-propanediol (1,2-PDO) at 125 °C, and gamma-valerolactone (GVL) at 160 °C. Based on these observations, the steps needed to achieve selective dissolution of the different PAs in the physical mixture were: PA66/6 dissolution in 1,2-propanediol at 125 °C, PA6 dissolution in DMSO at 145 °C, and PA66 dissolution in formic acid between 60 and 90 °C, according to experiments and solubility predictions. The solubility predictions for these steps indicated preferential dissolution for each PA (Table 1). The last polymer in the 10-component physical mixture was PET, for which solvents like GVL, NMP, and DMSO have been used for its dissolution in the literature (Chen et al., 2021; Poulakis and Papaspyrides, 2001; Walker et al., 2020). To ensure optimal selectivity, the PET was dissolved before PA6 and PA66, at 160 °C in GVL.

After the solvent selection for each step was completed, the overall procedure was demonstrated with the 10-component physical mixture of pure polymer pellets. Table 2 shows the results for the recovery of the polymers after each step in STRAP. Ideally, polymer precipitation should be achieved via a reduction in temperature since this has economic and environmental benefits (Munguía-López et al., 2023; Sánchez-Rivera et al., 2021). However, both PVC and PS required the addition of an antisolvent for their respective precipitation after dissolution. The recovery was 89 wt% or higher for each polymer, confirming the separation of one component per step. This was further confirmed with the FTIR of each recovered fraction, which was for the most part comparable to the pure polymer before STRAP (Fig. 3). As seen in Fig. 3(h), the only polymer that presented possible cross contamination after STRAP was PET. Additional peaks were detected after 3000 cm⁻¹, one of these pertaining to an N–H stretch at around 3300 cm⁻¹, which is characteristic of PA (Rodríguez et al., 2021). These peaks indicate contamination of the PET with PA66/6 residue from the previous step, since that polymer is also soluble in GVL at 160 °C. The rest of the recovered polymers showed comparable IR signatures to the corresponding pure polymers.

The thermal properties of each resin were analyzed with DSC to determine any changes after their recovery by STRAP. As seen in Table 3, the melting ($T_{m,2}$) and crystallization temperatures ($T_{c,2}$) of most of the polymers before and after STRAP were comparable. One noticeable difference was found in the melting temperature of the PET resin after STRAP. This could be due to some contamination from PA 66/6. Similarly, there were some differences in the values for the PAs, indicating possible contamination between them. In our previous studies, STRAP did not substantially affect the thermal properties of the recovered polymers (Sánchez-Rivera et al., 2023).

Table 2

STRAP steps for the recovery of 10 common packaging polymers from a physical mixture, with an initial amount of 1 g for each polymer and 30 g of the corresponding solvent.

Step	Polymer	Solvent	Temperature (°C)	Dissolution time (hr)	Antisolvent	Recovery (wt%)
1	PS	Toluene	35	0.5	IPA	89.06 %
2	PVC	THF	67	0.5	IPA	93.16 %
3	LDPE	O-xylene	80	0.5	None	90.84 %
4	HDPE	O-xylene	95	0.5	None	92.30 %
5	PP	O-xylene	115	0.5	None	97.61 %
6	EVOH	60 % DMSO-40 % water (v/v)	95	1	None	90.60 %
7	PA66/6	1,2-PDO	125	1	None	99.30 %
8	PET	GVL	160	1	None	106.72 %**
9	PA6	DMSO	145	1	None	99.13 %
10	PA66*	/	/	/	None	97.03 %

* PA66 was recovered as residue from the physical mixture.

** Solvent retention.

Table 3

Thermal properties of the polymers from a physical mixture before and after STRAP.

Step	Polymer	Before STRAP		After STRAP	
		$T_{m,2}$ (°C)	$T_{c,2}$ (°C)	$T_{m,2}$ (°C)	$T_{c,2}$ (°C)
1	PS	107.29	103.20	110.76	108.25
2	PVC	86.19	82.16	90.40	82.42
3	LDPE	111.24	99.73	110.89	97.58
4	HDPE	128.4	111.57	127.02	114.68
5	PP	163.51	116.94	160.71	115.23
6	EVOH	183.42	153.56	178.92	154.27
7	PA66/6	188.72	123.21	185.15	137.87
8	PET	242.21	154.31	221.91	145.91
9	PA6	221.76	159.49	212.24	169.78
10	PA66	258.98	213.70	260.97	228.70

3.3. Demonstration of STRAP with the post-industrial mixed plastic waste (MPW)

The 10 dissolution steps that were developed with the physical mixture targeting PS, PVC, LDPE, HDPE, PP, EVOH, PA66/6, PET, PA6, and PA66, were applied to the post-industrial MPW of interest. For the recovery of PA66, water was used as an antisolvent after its dissolution in formic acid. Fig. 4 shows the yields of each fraction recovered from the 10 STRAP steps. Majority components were LDPE, HDPE, and PET.

The fractions recovered in each STRAP step from the MPW were analyzed with ATR-FTIR to confirm the polymer separations. The majority components, LDPE, HDPE, and PET had comparable IR signatures to their corresponding virgin polymers (Fig. S1). Similar results were observed for EVOH and PA66/6. Other minor solid fractions recovered, PP and PVC, showed the presence of PE (Fig. 5(a,b)). Since there was no evidence of a high yield of PP or PVC and the IR did not confirm their recovery, these steps can be avoided for the MPW in this analysis. In the last two steps for the recovery of PA6 and PA66, potential PET residue was detected (Fig. 5(c,d)). This indicates further optimization of the PET dissolution step is required, via a change in temperature or time, to avoid cross contamination with this polymer. Overall, it would be beneficial to only focus on performing the processing steps for the majority polymers, LDPE, HDPE, and PET, in the presence of the remaining polymers.

The thermal properties of the fractions recovered from the MPW were analyzed and compared against the virgin resins used in the physical mixture. The LDPE, HDPE, EVOH, and PET samples after STRAP for the most part had comparable melting ($T_{m,2}$) and crystallization temperatures ($T_{c,2}$) to the pure resins (Table S2). However, the PP component had values indicating the presence of HDPE. This is different from the experiments done with the virgin resins and is likely because of differences in molecular weight of the PP and PE materials. This is an important property to consider in polymer dissolution since an increase in molecular weight decreases dissolution rates (Miller-Chou and

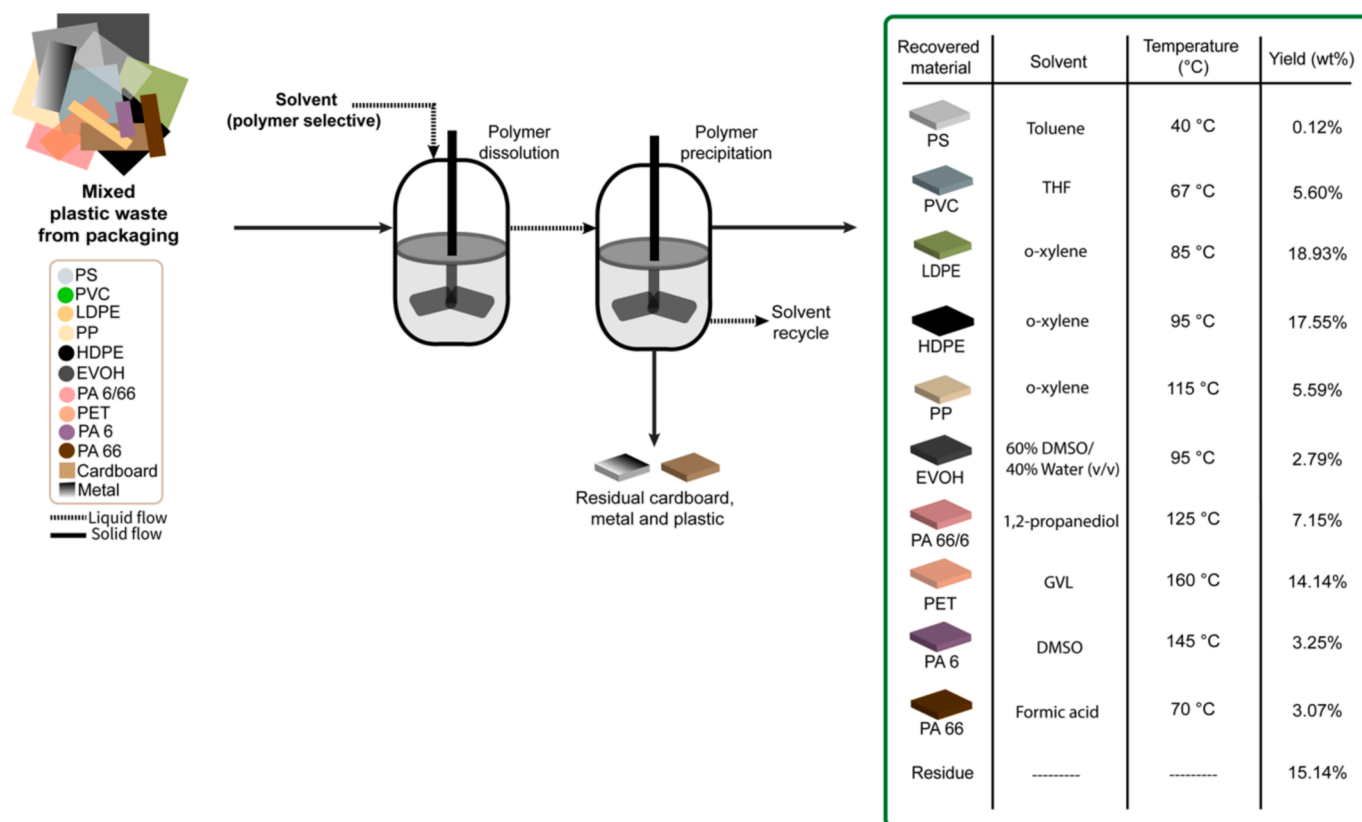


Fig. 4. Solvent-targeted recovery and precipitation (STRAP) with post-industrial mixed plastic waste (MPW) from packaging waste that includes multilayer films.

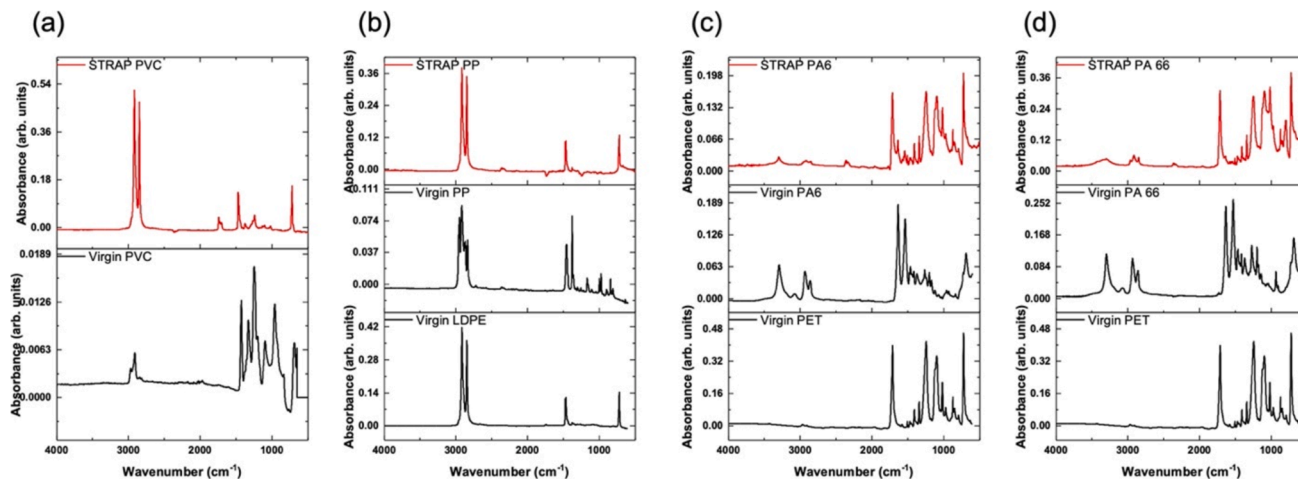


Fig. 5. IR spectra of the recovered polymers from MPW (in red) and the virgin polymers (in black): (a) PVC, (b) PP, (c) PA6, and (d) PA66. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Koenig, 2003). Tougher polymers with higher molecular weight distributions will require elevated temperatures for their proper dissolution in STRAP, and if multiple polymers have comparable molecular weights, their selective dissolutions can be challenging. More research is needed to better understand how to separate polyolefins like PP and PE from each other. The PA66 and PA6 recovered from the MPW showed thermal values indicating the presence of PET, which supports our results in the IR analysis. There is a possibility that the PAs were not present in the MPW, considering their low yield and differences in the IR and thermal properties. The PVC recovered from the MPW did not have values comparable to the PVC pure resin, possibly due to the presence of a PE impurity, as was observed in the IR spectra.

4. Conclusions

Selective polymer dissolution can be an effective way to recover different components from plastic waste mixtures. This was demonstrated by STRAP with physical mixtures containing PVC, PS, LDPE, HDPE, PP, EVOH, PET, PA6, PA66, and PA66/6, in which a sequence of steps were developed for their individual recovery. Our computational tools allowed us to predict solubility values for the targeted polymers and guide our solvent selection and experimental conditions for each step. This process was then demonstrated with a post-industrial mixed plastic waste (MPW) feed in which solids were recovered in each step of the procedure, with LDPE, HDPE, and PET representing the majority.

This method can be used to determine approximate compositions of the plastic waste feed and reduce the number of steps to only recover the main components. In an industrial scale process, minimizing the number of steps and solvents is of interest, considering needs for equipment, polymer drying, and solvent recovery. Additionally, it can be used to separate polymers of interest as a pretreatment for other recycling technologies. Future work will focus on applying this methodology to post-consumer mixed plastic waste.

CRedit authorship contribution statement

Kevin L. Sánchez-Rivera: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Panzheng Zhou:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Elizaveta Radkevich:** Writing – review & editing, Visualization, Investigation. **Anisha Sharma:** Writing – review & editing, Investigation. **Ezra Bar-Ziv:** Writing – review & editing, Investigation. **Reid C. Van Lehn:** Writing – review & editing, Investigation. **George W. Huber:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2025.01.022>.

Data availability

Data will be made available on request.

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